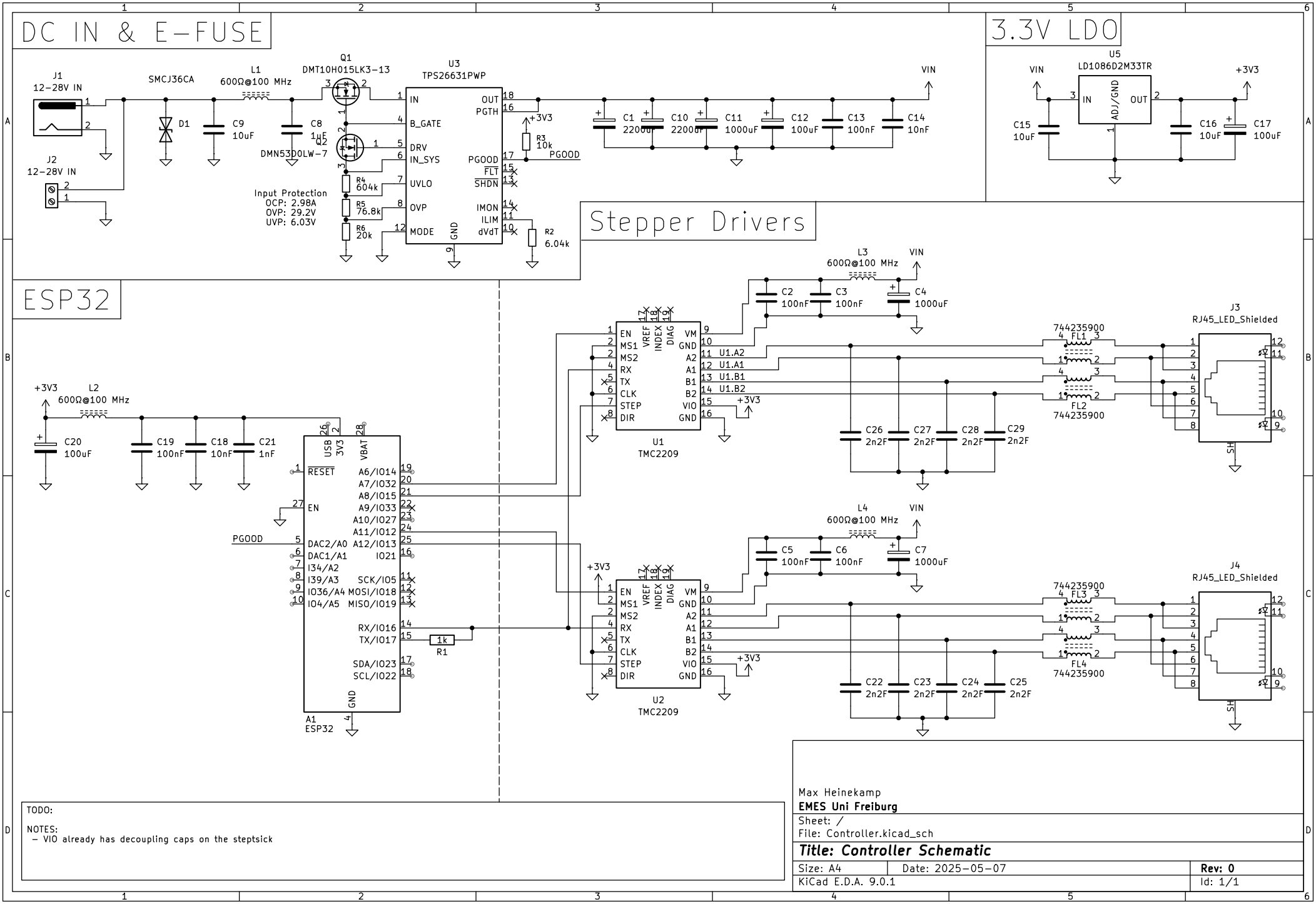

DC IN & E-FUSE
3.3V LDO
Stepper Drivers
ESP32
J1
12-28V IN
J2
12-28V IN
SMCJ36CA
D1
L1
600Ω@100 MHz
C9
10uF
C8
1uF
Q1
DMT10H015LK3-13
Q2
DMN53D0LW-7
U3
TPS26631PWP
IN
B_GATE
DRV
IN_SYS
UVLO
OVP
MODE
GND
OUT
PGTH
PGOOD
FLT
SHDN
IMON
ILIM
dVdT
R4
604k
R5
76.8k
R6
20k
R2
6.04k
R3
10k
+3V3
PGOOD
VIN
C1
2200uF
C10
2200uF
C11
1000uF
C12
100uF
C13
100nF
C14
10nF
Input Protection
OCP: 2.98A
OVP: 29.2V
UVP: 6.03V
U5
LD1086D2M33TR
IN
ADJ/GND
OUT
C15
10uF
C16
10uF
C17
100uF
L3
600Ω@100 MHz
C2
100nF
C3
100nF
C4
1000uF
U1
TMC2209
EN
MS1
MS2
RX
TX
CLK
STEP
DIR
VREF
INDEX
DIAG
VM
GND
A2
A1
B1
B2
VIO
U1.A2
U1.A1
U1.B1
U1.B2
C26
2n2F
C27
2n2F
C28
2n2F
C29
2n2F
744235900
FL1
FL2
J3
RJ45_LED_Shielded
L4
600Ω@100 MHz
C5
100nF
C6
100nF
C7
1000uF
U2
TMC2209
C22
2n2F
C23
2n2F
C24
2n2F
C25
2n2F
FL3
FL4
J4
RJ45_LED_Shielded
+3V3
L2
600Ω@100 MHz
C20
100uF
C19
100nF
C18
10nF
C21
1nF
A1
ESP32
RESET
EN
PGOOD
USB
3V3
VBAT
A6/IO14
A7/IO32
A8/IO15
A9/IO33
A10/IO27
A11/IO12
A12/IO13
IO21
I34/A2
I39/A3
IO36/A4
IO4/A5
SCK/IO5
MOSI/IO18
MISO/IO19
RX/IO16
TX/IO17
SDA/IO23
SCL/IO22
1k
R1
TODO:
NOTES:
– VIO already has decoupling caps on the steptsick
Max Heinekamp
EMES Uni Freiburg
Sheet: /
File: Controller.kicad_sch
Title: Controller Schematic
Size: A4
Date: 2025-05-07
Rev: 0
KiCad E.D.A. 9.0.1
Id: 1/1